```ts
export const SOFTWARE_NAME = '天平实验训练示例';
export const VERSION = 'V1.0';
export const OWNER = '示例著作权人';
export const STEP_NAMES = [
  '检查实验器材',
  '游码归零',
  '空载调平',
  '左盘放置物体',
  '由大到小添加砝码',
  '游码微调平衡',
  '读取并记录质量',
  '整理器材归位',
];
export const STEP_POINTS = [10, 10, 15, 15, 20, 10, 10, 10];
export const WEIGHTS = [500, 200, 100, 50]; // 0.1 g units
export interface Step {
  name: string;
  max: number;
  passed: boolean;
  score: number;
}
export interface Event {
  at: string;
  action: string;
  message: string;
  violation: boolean;
}
export interface Session {
  id: string;
  label: string;
  source: 'practice' | 'demo';
  createdAt: string;
  updatedAt: string;
  finishedAt: string | null;
  status: 'active' | 'finished';
  checked: boolean;
  calibrated: boolean;
  rider: number;
  nut: number;
  objectSide: 'box' | 'left' | 'right';
  objectMass: number;
  weights: { mass: number; side: 'left' | 'right' }[];
  tool: 'tweezers' | 'hand';
  steps: Step[];
  events: Event[];
  reading: number | null;
  score: number;
  feedback: string;
  balanced: boolean;
  measurement?: { weightTotal: number; rider: number; value: number; at: string };
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```
  difference: number;
}
export type Action =
  | { type: 'check' }
  | { type: 'rider'; value: number }
  | { type: 'nut'; value: number }
  | { type: 'tool'; value: 'tweezers' | 'hand' }
  | { type: 'object'; side: 'box' | 'left' | 'right' }
  | { type: 'addWeight'; mass: number; side: 'left' | 'right' }
  | { type: 'removeWeight'; index: number }
  | { type: 'read'; value: number }
  | { type: 'tidy' };
export interface ApiError {
  error: string;
}
import { createServer as httpServer } from 'node:http';
import type { IncomingMessage, ServerResponse } from 'node:http';
import { readFile } from 'node:fs/promises';
import { resolve, extname, sep } from 'node:path';
import { SOFTWARE_NAME, VERSION } from '../shared/types.ts';
import {
  applyAction,
  createSession,
  finishSession,
  InputError,
  validateAction,
} from './engine.ts';
import { Store } from './storage.ts';
class HttpError extends Error {
  status: number;
  constructor(status: number, message: string) {
    super(message);
    this.status = status;
  }
}
function json(res: ServerResponse, status: number, value: unknown) {
  res.writeHead(status, {
    'Content-Type': 'application/json; charset=utf-8',
    'Cache-Control': 'no-store',
  });
  res.end(JSON.stringify(value));
}
async function body(req: IncomingMessage): Promise<unknown> {
  if (!(req.headers['content-type'] || '').startsWith('application/json'))
    throw new HttpError(415, '请使用 application/json 请求。');
  let length = 0;
  const chunks: Buffer[] = [];
  for await (const chunk of req) {
    length += chunk.length;
    if (length > 16384) throw new HttpError(413, '请求体过大。');
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    chunks.push(Buffer.from(chunk));
  }
  try {
    return JSON.parse(Buffer.concat(chunks).toString('utf8'));
  } catch {
    throw new HttpError(400, '请求 JSON 格式错误。');
  }
}
export function createServer(dataDir: string, distDir: string) {
  const store = new Store(dataDir);
  const root = resolve(distDir);
  return httpServer(async (req, res) => {
    res.setHeader('X-Content-Type-Options', 'nosniff');
    res.setHeader('X-Frame-Options', 'DENY');
    res.setHeader('Referrer-Policy', 'no-referrer');
    try {
      const address = req.socket.localPort;
      const allowed = [`127.0.0.1:${address}`, `localhost:${address}`];
      if (!allowed.includes(req.headers.host || ''))
        throw new HttpError(403, '仅允许本机访问。');
      if (req.method !== 'GET' && req.method !== 'HEAD') {
        if (
          req.headers.origin &&
          !allowed.map((host) => `http://${host}`).includes(req.headers.origin)
        )
          throw new HttpError(403, '不允许其他网站修改本机实验记录。');
        if (req.headers['sec-fetch-site'] === 'cross-site')
          throw new HttpError(403, '拒绝跨站请求。');
      }
      const url = new URL(req.url || '/', `http://${req.headers.host}`);
      const path = url.pathname;
      if (path === '/api/health' && req.method === 'GET') {
        await store.list();
        return json(res, 200, { name: SOFTWARE_NAME, version: VERSION, status: 'ok' });
      }
      if (path === '/api/sessions' && req.method === 'GET')
        return json(res, 200, (await store.list()).reverse());
      if (path === '/api/sessions' && req.method === 'POST') {
        const input = (await body(req)) as { label?: unknown; source?: unknown };
        if (
          !input ||
          typeof input !== 'object' ||
          (input.label !== undefined &&
            (typeof input.label !== 'string' || input.label.trim().length > 80)) ||
          (input.source !== undefined &&
            (typeof input.source !== 'string' || !['practice', 'demo'].includes(input.source)))
        )
          throw new HttpError(400, '实验名称最多 80 字，来源须为 practice 或 demo。');
        const created = await store.mutate((sessions) => {
          const s = createSession(
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            typeof input.label === 'string' && input.label.trim()
              ? input.label.trim()
              : undefined,
            input.source as 'practice' | 'demo' | undefined,
          );
          sessions.push(s);
          return s;
        });
        return json(res, 201, created);
      }
      const match = path.match(
        /^\/api\/sessions\/([\w-]+)(?:\/(actions|finish|export))?$/,
      );
      if (match) {
        const [, id, operation] = match;
        if (req.method === 'GET' && (!operation || operation === 'export')) {
          const session = (await store.list()).find((s) => s.id === id);
          if (!session) throw new HttpError(404, '实验记录不存在。');
          if (operation === 'export')
            res.setHeader(
              'Content-Disposition',
              `attachment; filename="experiment-${id}.json"`,
            );
          return json(res, 200, session);
        }
        if (req.method === 'POST' && ['actions', 'finish'].includes(operation)) {
          const input = await body(req);
          if (operation === 'actions') validateAction(input);
          const session = await store.mutate((sessions) => {
            const index = sessions.findIndex((s) => s.id === id);
            if (index === -1) throw new HttpError(404, '实验记录不存在。');
            if (operation === 'actions') {
              validateAction(input);
              sessions[index] = applyAction(sessions[index], input);
            } else sessions[index] = finishSession(sessions[index]);
            return sessions[index];
          });
          return json(res, 200, session);
        }
        throw new HttpError(405, '不支持此请求方法。');
      }
      if (path.startsWith('/api/')) throw new HttpError(404, '接口不存在。');
      if (!['GET', 'HEAD'].includes(req.method || ''))
        throw new HttpError(405, '不支持此请求方法。');
      let decoded: string;
      try {
        decoded = decodeURIComponent(path);
      } catch {
        throw new HttpError(400, '路径编码无效。');
      }
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      if (decoded.includes('\0') || decoded.includes('\\'))
        throw new HttpError(400, '路径无效。');
      let file = resolve(root, `.${decoded}`);
      if (file !== root && !file.startsWith(root + sep))
        throw new HttpError(403, '禁止访问此路径。');
      if (decoded === '/' || !extname(decoded)) file = resolve(root, 'index.html');
      let content: Buffer;
      try {
        content = await readFile(file);
      } catch {
        throw new HttpError(404, '页面不存在，请先运行构建脚本。');
      }
      const mime: Record<string, string> = {
        '.html': 'text/html; charset=utf-8',
        '.js': 'text/javascript; charset=utf-8',
        '.css': 'text/css; charset=utf-8',
        '.svg': 'image/svg+xml',
        '.png': 'image/png',
        '.ico': 'image/x-icon',
      };
      res.writeHead(200, {
        'Content-Type': mime[extname(file)] || 'application/octet-stream',
        'Cache-Control': 'no-cache',
      });
      res.end(req.method === 'HEAD' ? undefined : content);
    } catch (error) {
      const status =
        error instanceof HttpError
          ? error.status
          : error instanceof InputError
            ? 400
            : 500;
      if (status === 500) console.error('实验数据服务异常：', error);
      if (!res.headersSent)
        json(res, status, {
          error:
            status === 500
              ? '读取或保存记录失败；原文件已保留，请检查数据目录或恢复备份。'
              : (error as Error).message,
        });
      else res.end();
    }
  });
}
import { randomUUID } from 'node:crypto';
import { STEP_NAMES, STEP_POINTS, WEIGHTS } from '../shared/types.ts';
import type { Action, Session } from '../shared/types.ts';
export class InputError extends Error {}
export function createSession(
  label = '天平实验练习',
```

```
  source: 'practice' | 'demo' = 'practice',
): Session {
  const now = new Date().toISOString();
  return {
    id: randomUUID(),
    label,
    source,
    createdAt: now,
    updatedAt: now,
    finishedAt: null,
    status: 'active',
    checked: false,
    calibrated: false,
    rider: 10,
    nut: 2,
    objectSide: 'box',
    objectMass: 374,
    weights: [],
    tool: 'tweezers',
    steps: STEP_NAMES.map((name, i) => ({
      name,
      max: STEP_POINTS[i],
      passed: false,
      score: 0,
    })),
    events: [],
    reading: null,
    score: 0,
    feedback: '请先检查器材，再将游码归零。',
    balanced: false,
    difference: 10,
  };
}
function integer(value: unknown, min: number, max: number): value is number {
  return Number.isInteger(value) && Number(value) >= min && Number(value) <= max;
}
export function validateAction(value: unknown): asserts value is Action {
  if (!value || typeof value !== 'object')
    throw new InputError('操作必须是 JSON 对象。');
  const a = value as Record<string, unknown>;
  if (typeof a.type !== 'string') throw new InputError('操作类型必须是字符串。');
  const valid =
    ['check', 'tidy'].includes(a.type) ||
    (a.type === 'rider' && integer(a.value, 0, 50)) ||
    (a.type === 'nut' && integer(a.value, -5, 5)) ||
    (a.type === 'tool' && typeof a.value === 'string' && ['tweezers', 'hand'].includes(a.value)) ||
    (a.type === 'object' && typeof a.side === 'string' && ['box', 'left', 'right'].includes(a.side)) ||
    (a.type === 'addWeight' &&
      WEIGHTS.includes(Number(a.mass)) &&
      typeof a.mass === 'number' &&
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      typeof a.side === 'string' && ['left', 'right'].includes(a.side)) ||
    (a.type === 'removeWeight' && integer(a.index, 0, 3)) ||
    (a.type === 'read' && integer(a.value, 0, 10000));
  if (!valid) throw new InputError('操作类型或参数无效。质量参数使用整数十分之一克。');
}
function pass(s: Session, index: number) {
  if (s.steps.slice(0, index).every((step) => step.passed)) {
    s.steps[index].passed = true;
    s.steps[index].score = s.steps[index].max;
  }
}
function empty(s: Session) {
  return s.objectSide === 'box' && s.weights.length === 0;
}
function measure(s: Session) {
  const left =
    (s.objectSide === 'left' ? s.objectMass : 0) +
    s.weights.filter((w) => w.side === 'left').reduce((n, w) => n + w.mass, 0);
  const right =
    (s.objectSide === 'right' ? s.objectMass : 0) +
    s.weights.filter((w) => w.side === 'right').reduce((n, w) => n + w.mass, 0);
  s.difference = left - right - s.rider + s.nut * 10;
  s.balanced = s.difference === 0;
}
function properWeights(s: Session) {
  return (
    s.weights.length > 0 &&
    s.weights.every((w) => {
      const event = [...s.events]
        .reverse()
        .find((e) => e.action.startsWith(`addWeight:${w.mass}:`));
      return w.side === 'right' && event?.action.endsWith(':valid');
    })
  );
}
function readyToRead(s: Session) {
  return (
    s.calibrated &&
    s.nut === 0 &&
    s.objectSide === 'left' &&
    properWeights(s) &&
    s.balanced &&
    s.steps[5].passed
  );
}
export function applyAction(previous: Session, action: Action): Session {
  validateAction(action);
  if (previous.status !== 'active') throw new InputError('实验已结束，请新建实验。');
  const s = structuredClone(previous);
  let violation = false;
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  let message = '';
  let tag: string = action.type;
  const reject = (text: string) => {
    violation = true;
    message = text;
  };
  switch (action.type) {
    case 'check':
      s.checked = true;
      pass(s, 0);
      message = '器材检查完成：天平、物体、砝码、镊子齐全。';
      break;
    case 'rider':
      s.rider = action.value;
      message = `游码已移至 ${(s.rider / 10).toFixed(1)} g。`;
      if (!s.checked) reject('请先检查器材。');
      else if (s.rider === 0 && empty(s)) pass(s, 1);
      else if (!s.calibrated) reject('请将游码归零并完成空载调平。');
      break;
    case 'nut':
      s.nut = action.value;
      s.calibrated = false;
      message = `平衡螺母设置为 ${s.nut}。`;
      if (!empty(s))
        reject('有负载时禁止调节平衡螺母；请取下物体和砝码、游码归零后重新调平。');
      else if (!s.checked || !s.steps[1].passed || s.rider !== 0)
        reject('调平前请先检查器材并将游码归零。');
      else if (s.nut === 0) {
        s.calibrated = true;
        pass(s, 2);
        message = '空载调平完成。';
      }
      break;
    case 'tool':
      s.tool = action.value;
      message =
        s.tool === 'tweezers'
          ? '已选用镊子。'
          : '已切换为徒手操作；取放砝码应使用镊子。';
      break;
    case 'object':
      s.objectSide = action.side;
      message =
        action.side === 'box'
          ? '物体已放回器材区。'
          : `物体已放在 ${action.side === 'left' ? '左' : '右'} 盘。`;
      if (action.side === 'right') reject('应左物右码，请将物体放在左盘。');
      else if (
        action.side === 'left' &&
        (!s.calibrated || s.nut !== 0 || s.rider !== 0)
```

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    )
      reject('放物前请游码归零，并完成空载调平。');
    else if (action.side === 'left') pass(s, 3);
    break;
  case 'addWeight': {
    if (s.weights.some((w) => w.mass === action.mass))
      throw new InputError('该砝码已在托盘中。');
    const last = s.weights.at(-1);
    const valid =
      s.tool === 'tweezers' &&
      action.side === 'right' &&
      s.objectSide === 'left' &&
      s.calibrated &&
      s.nut === 0 &&
      s.steps[3].passed &&
      (!last || action.mass < last.mass);
    s.weights.push({ mass: action.mass, side: action.side });
    tag = `addWeight:${action.mass}:${valid ? 'valid' : 'invalid'}`;
    message = `已添加 ${(action.mass / 10).toFixed(1)} g 砝码。`;
    if (!valid)
      reject(
        '请先正确调平并左盘放物，使用镊子向右盘按从大到小顺序加码；错误砝码请取下后重新添加。',
      );
    else pass(s, 4);
    break;
  }
  case 'removeWeight': {
    if (!s.weights[action.index]) throw new InputError('砝码索引不存在。');
    const [removed] = s.weights.splice(action.index, 1);
    message = `已取回 ${(removed.mass / 10).toFixed(1)} g 砝码。`;
    if (s.tool !== 'tweezers') reject('取回砝码也应使用镊子。');
    break;
  }
  case 'read':
    measure(s);
    if (!readyToRead(s))
      reject('当前不能读数：需正确调平、左物右码、规范取码，并完成游码微调平衡。');
    else if (action.value !== s.objectMass)
      reject('读数不正确，请将右盘砝码总质量与游码示值相加。');
    else {
      s.reading = action.value;
      s.measurement = {
        weightTotal: s.weights.reduce((n, w) => n + w.mass, 0),
        rider: s.rider,
        value: action.value,
        at: new Date().toISOString(),
      };
      pass(s, 6);
      message = `测量结果 ${(s.reading / 10).toFixed(1)} g，记录成功。`;
    }
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      break;
    case 'tidy':
      if (!s.steps[6].passed) reject('请先完成正确读数，再整理器材。');
      else {
        s.objectSide = 'box';
        s.weights = [];
        s.rider = 0;
        s.nut = 0;
        s.tool = 'tweezers';
        s.calibrated = true;
        pass(s, 7);
        message = '物体和砝码已归位、游码已归零，实验完成。';
      }
      break;
  }
  s.updatedAt = new Date().toISOString();
  s.events.push({ at: s.updatedAt, action: tag, message, violation });
  measure(s);
  if (
    action.type === 'rider' &&
    !violation &&
    s.rider > 0 &&
    s.calibrated &&
    s.nut === 0 &&
    s.objectSide === 'left' &&
    properWeights(s) &&
    s.balanced
  ) {
    pass(s, 5);
    message = '游码微调完成，天平平衡。请读取右盘砝码与游码的合计质量。';
    s.events[s.events.length - 1].message = message;
  }
  s.score = s.steps.reduce((n, step) => n + step.score, 0);
  s.feedback = message;
  return s;
}
export function finishSession(previous: Session): Session {
  if (previous.status === 'finished') return previous;
  const s = structuredClone(previous);
  s.status = 'finished';
  s.finishedAt = new Date().toISOString();
  s.updatedAt = s.finishedAt;
  s.feedback = `实验已归档，练习完成得分 ${s.score} / 100。`;
  s.events.push({
    at: s.finishedAt,
    action: 'finish',
    message: s.feedback,
    violation: false,
  });
  return s;
```

```
}
import { resolve } from 'node:path';
import { fileURLToPath } from 'node:url';
import { createServer } from './app.ts';
import { SOFTWARE_NAME, VERSION } from '../shared/types.ts';
const root = fileURLToPath(new URL('../', import.meta.url));
const port = Number(process.env.PORT || 3187);
if (!Number.isInteger(port) || port < 1 || port > 65535)
  throw new Error('PORT 必须为 1 至 65535 的整数。');
const server = createServer(
  resolve(root, process.env.DATA_DIR || 'data'),
  resolve(root, 'dist'),
);
server.on('error', (error: NodeJS.ErrnoException) => {
  console.error(
    error.code === 'EADDRINUSE'
      ? `端口 ${port} 已被占用，请关闭旧服务或修改 PORT 后重试。`
      : error.message,
  );
  process.exitCode = 1;
});
server.listen(port, '127.0.0.1', () =>
  console.log(`${SOFTWARE_NAME} ${VERSION}\n 本机访问：http://127.0.0.1:${port}`),
);
for (const signal of ['SIGINT', 'SIGTERM'] as const)
  process.on(signal, () => server.close(() => process.exit(0)));
import { mkdir, readFile, rename, writeFile } from 'node:fs/promises';
import { join } from 'node:path';
import { randomUUID } from 'node:crypto';
import { STEP_POINTS, WEIGHTS } from '../shared/types.ts';
import type { Session } from '../shared/types.ts';
function validSession(value: unknown): value is Session {
  if (!value || typeof value !== 'object') return false;
  const s = value as Session;
  return (
    typeof s.id === 'string' &&
    typeof s.label === 'string' &&
    ['practice', 'demo'].includes(s.source) &&
    ['active', 'finished'].includes(s.status) &&
    typeof s.createdAt === 'string' &&
    typeof s.updatedAt === 'string' &&
    (s.finishedAt === null || typeof s.finishedAt === 'string') &&
    typeof s.checked === 'boolean' &&
    typeof s.calibrated === 'boolean' &&
    Number.isInteger(s.rider) &&
    s.rider >= 0 &&
    s.rider <= 50 &&
    Number.isInteger(s.nut) &&
    s.nut >= -5 &&
    s.nut <= 5 &&
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    s.objectMass === 374 &&
    ['box', 'left', 'right'].includes(s.objectSide) &&
    ['tweezers', 'hand'].includes(s.tool) &&
    Array.isArray(s.weights) &&
    s.weights.length <= WEIGHTS.length &&
    new Set(s.weights.map((w) => w?.mass)).size === s.weights.length &&
    s.weights.every(
      (w) => w && WEIGHTS.includes(w.mass) && ['left', 'right'].includes(w.side),
    ) &&
    Array.isArray(s.steps) &&
    s.steps.length === 8 &&
    s.steps.every(
      (step, i) =>
        step &&
        typeof step.name === 'string' &&
        typeof step.passed === 'boolean' &&
        step.max === STEP_POINTS[i] &&
        step.score === (step.passed ? step.max : 0),
    ) &&
    s.score === s.steps.reduce((total, step) => total + step.score, 0) &&
    Array.isArray(s.events) &&
    s.events.every(
      (e) =>
        e &&
        typeof e.at === 'string' &&
        typeof e.action === 'string' &&
        typeof e.message === 'string' &&
        typeof e.violation === 'boolean',
    ) &&
    (s.reading === null || Number.isInteger(s.reading)) &&
    typeof s.feedback === 'string' &&
    typeof s.balanced === 'boolean' &&
    Number.isInteger(s.difference)
  );
}
export class Store {
  private queue: Promise<unknown> = Promise.resolve();
  private file: string;
  private directory: string;
  constructor(directory: string) {
    this.directory = directory;
    this.file = join(directory, 'sessions.json');
  }
  private async read(): Promise<Session[]> {
    let raw: string;
    try {
      raw = await readFile(this.file, 'utf8');
    } catch (error) {
      if ((error as NodeJS.ErrnoException).code === 'ENOENT') return [];
      throw error;
```

```
    }
    const sessions: unknown = JSON.parse(raw);
    if (
      !Array.isArray(sessions) ||
      !sessions.every(validSession) ||
      new Set(sessions.map((s) => s.id)).size !== sessions.length
    ) {
      throw new Error('记录文件损坏；请保留原文件并从备份恢复。');
    }
    return sessions as Session[];
  }
  async list(): Promise<Session[]> {
    await this.queue;
    return this.read();
  }
  mutate<T>(change: (sessions: Session[]) => T): Promise<T> {
    const operation = this.queue.then(async () => {
      const sessions = await this.read();
      const result = change(sessions);
      await mkdir(this.directory, { recursive: true });
      const temporary = `${this.file}.${randomUUID()}.tmp`;
      await writeFile(temporary, JSON.stringify(sessions, null, 2), {
        encoding: 'utf8',
        flag: 'wx',
      });
      await rename(temporary, this.file);
      return result;
    });
    this.queue = operation.catch(() => undefined);
    return operation;
  }
}
import React, { useEffect, useState } from 'react';
import { createRoot } from 'react-dom/client';
import {
  SOFTWARE_NAME,
  VERSION,
  STEP_NAMES,
  STEP_POINTS,
  WEIGHTS,
  type Session,
  type Action,
} from '../shared/types';
import './style.css';
const grams = (n: number) => (n / 10).toFixed(1);
const date = (s: string) => new Date(s).toLocaleString('zh-CN', { hour12: false });
async function api<T>(path: string, body?: unknown): Promise<T> {
  const response = await fetch(
    '/api' + path,
    body === undefined
```

```
      ? {}
      : {
          method: 'POST',
          headers: { 'Content-Type': 'application/json' },
          body: JSON.stringify(body),
        },
  );
  const data = await response.json();
  if (!response.ok) throw new Error(data.error || '操作未完成，请重试');
  return data;
}
function Balance({ session: s }: { session: Session }) {
  const angle = Math.max(-13, Math.min(13, -s.difference / 12));
  const pan = (side: 'left' | 'right', x: number) => {
    const y =
      175 + (side === 'left' ? -1 : 1) * Math.sin((angle * Math.PI) / 180) * 215;
    return (
      <g>
        <path
          d={`M ${x} ${y} L ${x - 66} ${y + 102} M ${x} ${y} L ${x + 66} ${y + 102}`}
          stroke="#8b9fa8"
          strokeWidth="2"
          fill="none"
        />
        <path
          d={`M ${x - 78} ${y + 102} Q ${x - 71} ${y + 123} ${x} ${y + 125} Q ${x + 71} ${y + 123} ${x +
          78} ${y + 102} Z`}
          fill="url(#steel)"
          stroke="#71909e"
        />
        {s.objectSide === side && (
          <g>
            <path
              d={`M ${x - 25} ${y + 54} l 38 -10 20 14 -1 43 -45 0 -13 -12 Z`}
              fill="#bb7658"
              stroke="#925339"
            />
            <path
              d={`M ${x - 25} ${y + 54} l 20 12 38 -8 M ${x - 5} ${y + 66} v 35`}
              fill="none"
              stroke="#e4ac87"
              strokeWidth="2"
            />
          </g>
        )}
        {s.weights
          .filter((w) => w.side === side)
          .map((w, i) => (
            <g
              key={w.mass}
              transform={`translate(${x - 48 + i * 32},${y + 81 - (w.mass === 500 ? 12 : 0)})`}
```

```
            >
              <rect x="7" y="-7" width="12" height="8" rx="3" fill="#8e7041" />
              <path
                d="M 6 0 L 20 0 L 25 20 L 1 20 Z"
                fill="url(#brass)"
                stroke="#957645"
              />
              <text x="13" y="14" textAnchor="middle" fontSize="9" fill="#4d3d25">
                {w.mass / 10}
              </text>
            </g>
          ))}
        <text x={x} y="335" textAnchor="middle" className="pan-label">
          {side === 'left' ? '左盘 • 放置物体' : '右盘 • 添加砝码'}
        </text>
      </g>
    );
  };
  return (
    <svg
      className="balance"
      viewBox="0 0 820 390"
      role="img"
      aria-label={`托盘天平，${s.balanced ? '已平衡' : '未平衡'}，左盘${s.objectSide === 'left' ? '有物体' :
      '无物体'}`}
    >
      <defs>
        <linearGradient id="steel" x1="0" y1="0" x2="0" y2="1">
          <stop stopColor="#ecf4f5" />
          <stop offset=".6" stopColor="#b9cbd2" />
          <stop offset="1" stopColor="#7c99a7" />
        </linearGradient>
        <linearGradient id="brass">
          <stop stopColor="#d5bb81" />
          <stop offset=".5" stopColor="#eee0b4" />
          <stop offset="1" stopColor="#ab8c4c" />
        </linearGradient>
        <linearGradient id="base">
          <stop stopColor="#265468" />
          <stop offset="1" stopColor="#153748" />
        </linearGradient>
      </defs>
      <ellipse cx="410" cy="364" rx="218" ry="10" fill="#173d4c" opacity=".07" />
      <path d="M 332 349 L 346 327 H 474 L 490 349 Z" fill="url(#base)" />
      <rect x="393" y="172" width="34" height="155" rx="4" fill="url(#steel)" />
      <path d="M 379 327 L 398 282 H 422 L 441 327" fill="#27556b" />
      <path d="M 357 177 A 56 56 0 0 1 463 177" fill="#fff" stroke="#b9cbd2" />
      {[-40, -20, 0, 20, 40].map((v) => (
        <line
          key={v}
          x1={410 + v}
```

```
          y1={v === 0 ? 124 : 130 + Math.abs(v) / 4}
          x2={410 + v}
          y2={v === 0 ? 138 : 139 + Math.abs(v) / 4}
          stroke={v === 0 ? '#0d887f' : '#8ba2ad'}
          strokeWidth={v === 0 ? 3 : 1}
        />
      ))}
      <g transform={`rotate(${angle},410,175)`} className="beam">
        <path
          d="M 193 173 L 410 161 L 627 173 L 627 185 L 410 180 L 193 185 Z"
          fill="url(#steel)"
          stroke="#73909e"
        />
        <line x1="410" y1="171" x2="410" y2="124" stroke="#148d85" strokeWidth="3" />
        <circle cx="410" cy="175" r="9" fill="#31586a" />
        <circle cx="410" cy="175" r="3" fill="#e5eef1" />
        <rect x="624" y="167" width="18" height="24" rx="3" fill="url(#brass)" />
        <rect x="180" y="167" width="18" height="24" rx="3" fill="url(#brass)" />
        <rect
          x="426"
          y="186"
          width="181"
          height="15"
          rx="2"
          fill="#dce7eb"
          stroke="#8ea7b2"
        />
        {Array.from({ length: 26 }, (_, i) => (
          <line
            key={i}
            x1={430 + i * 6.8}
            x2={430 + i * 6.8}
            y1="187"
            y2={i % 5 === 0 ? 197 : 192}
            stroke="#637d89"
          />
        ))}
        <rect
          x={426 + s.rider * 3.4}
          y="182"
          width="9"
          height="25"
          rx="2"
          fill="#187e7b"
        />
      </g>
      {pan('left', 195)}
      {pan('right', 625)}
      <text
        x="410"
```

```
        y="349"
        textAnchor="middle"
        fill="#d8e7eb"
        fontSize="10"
        letterSpacing="3"
      >
        TRAY BALANCE
      </text>
      <g transform="translate(350,35)">
        <rect
          width="120"
          height="28"
          rx="14"
          fill={s.balanced ? '#d8eee7' : '#e1e9ee'}
        />
        <circle cx="17" cy="14" r="3" fill={s.balanced ? '#168778' : '#617a89'} />
        <text x="67" y="19" textAnchor="middle" fill="#285365" fontSize="12">
          {s.balanced ? '指针已平衡' : '指针未平衡'}
        </text>
      </g>
    </svg>
  );
}
function App() {
  const [tab, setTab] = useState('实验台'),
    [session, setSession] = useState<Session | null>(null),
    [history, setHistory] = useState<Session[]>([]),
    [busy, setBusy] = useState(false),
    [error, setError] = useState(''),
    [label, setLabel] = useState('天平质量测量练习'),
    [source, setSource] = useState<'practice' | 'demo'>('practice'),
    [side, setSide] = useState<'left' | 'right'>('right'),
    [reading, setReading] = useState(''),
    [report, setReport] = useState<Session | null>(null),
    [reset, setReset] = useState(false),
    [online, setOnline] = useState(false);
  const run = async (fn: () => Promise<void>) => {
    setBusy(true);
    setError('');
    try {
      await fn();
    } catch (e) {
      setError(e instanceof Error ? e.message : '连接失败，请检查本地服务后重试');
    } finally {
      setBusy(false);
    }
  };
  const load = () =>
    run(async () => {
      await api('/health');
```

```
      setOnline(true);
      setHistory(await api<Session[]>('/sessions'));
      const id = localStorage.getItem('balance-session');
      if (id) setSession(await api<Session>('/sessions/' + id));
    });
  useEffect(() => {
    void load();
  }, []);
  useEffect(() => {
    if (!reset && !report) return;
    const dialog = document.querySelector<HTMLElement>(
      report ? '.report[role="dialog"]' : '.new-dialog[role="dialog"]',
    );
    if (!dialog) return;
    const previousFocus = document.activeElement instanceof HTMLElement
      ? document.activeElement : null;
    const focusable = () => Array.from(dialog.querySelectorAll<HTMLElement>(
      'button:not(:disabled), input:not(:disabled), select:not(:disabled), a[href],
      [tabindex]:not([tabindex="-1"])',
    )).filter((element) => element.getClientRects().length > 0);
    dialog.tabIndex = -1;
    const focusFirst = () => (focusable()[0] ?? dialog).focus();
    focusFirst();
    const onKeyDown = (event: KeyboardEvent) => {
      if (event.key === 'Escape') {
        event.preventDefault();
        if (report) setReport(null);
        else setReset(false);
      } else if (event.key === 'Tab') {
        const items = focusable();
        const first = items[0], last = items[items.length - 1];
        if (!first) {
          event.preventDefault();
          dialog.focus();
        } else if (event.shiftKey &&
          (document.activeElement === first || !dialog.contains(document.activeElement))) {
          event.preventDefault();
          last.focus();
        } else if (!event.shiftKey &&
          (document.activeElement === last || !dialog.contains(document.activeElement))) {
          event.preventDefault();
          first.focus();
        }
      }
    };
    const onFocusIn = (event: FocusEvent) => {
      if (event.target instanceof Node && !dialog.contains(event.target)) focusFirst();
    };
    document.addEventListener('keydown', onKeyDown);
    document.addEventListener('focusin', onFocusIn);
    return () => {
```

```
      document.removeEventListener('keydown', onKeyDown);
      document.removeEventListener('focusin', onFocusIn);
      if (previousFocus?.isConnected) previousFocus.focus();
    };
  }, [reset, report?.id]);
  const create = () =>
    run(async () => {
      const s = await api<Session>('/sessions', {
        label: label.trim() || '天平质量测量练习',
        source,
      });
      setSession(s);
      localStorage.setItem('balance-session', s.id);
      setReading('');
      setSide('right');
      setReset(false);
      setTab('实验台');
      setHistory(await api('/sessions'));
    });
  const act = (action: Action) =>
    session &&
    run(async () =>
      setSession(await api<Session>(`/sessions/${session.id}/actions`, action)),
    );
  const finish = () =>
    session &&
    run(async () => {
      const s = await api<Session>(`/sessions/${session.id}/finish`, {});
      setSession(s);
      setReport(s);
      setHistory(await api('/sessions'));
    });
  const disabled = busy || session?.status === 'finished';
  const submitReading = () => {
    if (
      !/^\d+(\.\d)?$/.test(reading) ||
      Number(reading) < 0 ||
      Number(reading) > 1000
    ) {
      setError('读数需为 0 至 1000 g 的数字，最多保留一位小数。');
      return;
    }
    void act({ type: 'read', value: Math.round(Number(reading) * 10) });
  };
  const download = (s: Session) => {
    const a = document.createElement('a');
    a.href = `/api/sessions/${s.id}/export`;
    a.download = `天平实验-${s.id}.json`;
    a.click();
  };
```

```
  return (
    <>
      <header className="topbar">
        <div className="brand-icon">⚖</div>
        <div className="brand">
          <strong>{SOFTWARE_NAME}</strong>
          <span> 实验教学 ·交互练习与过程评测</span>
        </div>
        <span className="version">{VERSION}</span>
        <div className="connection">
          <i className={online ? 'on' : ''} />
          {online ? '本地服务已连接' : '正在连接本地服务'}
        </div>
      </header>
      <div className="navigation">
        <nav aria-label=" 主导航">
          {['实验台', '实验记录', '实验指南'].map((t, i) => (
            <button
              key={t}
              className={tab === t ? 'selected' : ''}
              onClick={() => {
                setTab(t);
                if (i === 1) void run(async () => setHistory(await api('/sessions')));
              }}
            >
              <span>{['◈', '▤', '◎'][i]}</span>
              {t}
            </button>
          ))}
        </nav>
        <span> 认识仪器，规范操作，准确测量。</span>
      </div>
      <main>
        {error && (
          <div role="alert" className="error">
            {error}
            <button onClick={() => void load()}> 重新连接</button>
          </div>
        )}
        {tab === '实验台' && (
          <>
            <div className="page-heading">
              <div>
                <div className="eyebrow"> 物理实验 / 质量测量</div>
                <h1> 把每一步，放在天平上。</h1>
                <p> 按顺序完成八个实验环节，操作过程将自动保存。</p>
              </div>
              <button className="secondary" onClick={() => setReset(true)}>
                ＋ 新建实验
              </button>
```

```
        </div>
        {!session ? (
          <section className="start-panel">
            <div className="start-mark">⚖</div>
            <h2> 开始你的第一轮测量</h2>
            <p> 检查器材、调整平衡，再测量物体的质量。</p>
            <label>
              实验名称
              <input
                value={label}
                onChange={(e) => setLabel(e.target.value)}
                maxLength={80}
              />
            </label>
            <label>
              记录来源
              <select
                value={source}
                onChange={(e) => setSource(e.target.value as 'practice' | 'demo')}
              >
                <option value="practice"> 自主练习</option>
                <option value="demo"> 演示记录</option>
              </select>
            </label>
            <button
              className="primary"
              disabled={busy}
              onClick={() => void create()}
            >
              开始实验 →
            </button>
          </section>
        ) : (
          <div className="workbench">
            <div className="left-column">
              <section className="instrument-panel">
                <div className="panel-heading">
                  <div>
                    <strong>{session.label}</strong>
                    <span className="badge">
                      {session.source === 'demo' ? '演示记录' : '自主练习'}
                    </span>
                    {session.status === 'finished' && (
                      <span className="badge"> 已归档</span>
                    )}
                  </div>
                  <span className="subtle"> 分度值 0.1 g</span>
                </div>
                <Balance session={session} />
                <div className="instrument-readings">
```

```
            <div>
              <span> 游码示值</span>
              <strong>
                {grams(session.rider)}
                <small>g</small>
              </strong>
            </div>
            <div>
              <span> 砝码总质量</span>
              <strong>
                {grams(session.weights.reduce((n, w) => n + w.mass, 0))}
                <small>g</small>
              </strong>
            </div>
            <div>
              <span> 空载调平</span>
              <strong className="status-word">
                {session.calibrated ? '已完成' : '待调整'}
              </strong>
            </div>
          </div>
        </section>
        <div
          className={`feedback ${session.events.at(-1)?.violation ? 'warning' : ''}`}
          role="status"
        >
          <span>{session.events.at(-1)?.violation ? '!' : 'i'}</span>
          <div>
            <strong>
              {session.events.at(-1)?.violation ? '操作提示' : '当前指引'}
            </strong>
            <p>{session.feedback}</p>
          </div>
        </div>
        <section className="controls">
          <div className="panel-heading">
            <strong> 器材与操作</strong>
            <span className="subtle"> 依次完成准备、测量与整理</span>
          </div>
          <fieldset disabled={disabled}>
            <div className="control-grid">
              <div className="control-group">
                <h3>
                  <b>01</b> 准备与调平
                </h3>
                <button
                  className={session.checked ? 'done-button' : 'secondary'}
                  onClick={() => void act({ type: 'check' })}
                >
                  {session.checked ? '✓ 器材检查完成' : '检查实验器材'}
```

```
        </button>
        <label className="slider-label">
          平衡螺母 <output>{session.nut}</output>
          <input
            aria-label=" 平衡螺母"
            type="range"
            min="-5"
            max="5"
            value={session.nut}
            onChange={(e) =>
              void act({ type: 'nut', value: Number(e.target.value) })
            }
          />
        </label>
        <small> 空载时，先游码归零，再调螺母至 0。</small>
      </div>
      <div className="control-group">
        <h3>
          <b>02</b> 放物与取码
        </h3>
        <label>
          取码工具
          <select
            aria-label=" 取码工具"
            value={session.tool}
            onChange={(e) =>
              void act({
                type: 'tool',
                value: e.target.value as 'hand' | 'tweezers',
              })
            }
          >
            <option value="tweezers"> 镊子</option>
            <option value="hand"> 徒手（不规范操作）</option>
          </select>
        </label>
        <label>
          物体位置
          <select
            aria-label=" 物体位置"
            value={session.objectSide}
            onChange={(e) =>
              void act({
                type: 'object',
                side: e.target.value as 'box' | 'left' | 'right',
              })
            }
          >
            <option value="box"> 器材盒</option>
            <option value="left"> 左盘</option>
```

```
            <option value="right"> 右盘</option>
          </select>
        </label>
        <label>
          砝码放置
          <select
            aria-label=" 砝码放置"
            value={side}
            onChange={(e) =>
              setSide(e.target.value as 'left' | 'right')
            }
          >
            <option value="right"> 右盘</option>
            <option value="left"> 左盘</option>
          </select>
        </label>
      </div>
      <div className="control-group">
        <h3>
          <b>03</b> 砝码与游码
        </h3>
        <div className="weight-buttons">
          {WEIGHTS.map((w) => (
            <button
              key={w}
              disabled={session.weights.some((x) => x.mass === w)}
              onClick={() =>
                void act({ type: 'addWeight', mass: w, side })
              }
              aria-label={`添加 ${w / 10} 克砝码`}
            >
              <span>▰</span>
              {w / 10}
              <small>g</small>
            </button>
          ))}
        </div>
        <div className="placed-weights">
          {session.weights.length ? (
            session.weights.map((w, i) => (
              <button
                key={w.mass}
                onClick={() =>
                  void act({ type: 'removeWeight', index: i })
                }
                aria-label={`取回 ${w.mass / 10} 克砝码`}
              >
                {w.mass / 10}g {w.side === 'left' ? '左' : '右'} ×
              </button>
            ))
```

```
            ) : (
              <span> 砝码均在器材盒内</span>
            )}
          </div>
          <label className="slider-label">
            游码 <output>{grams(session.rider)} g</output>
            <input
              aria-label=" 游码"
              type="range"
              min="0"
              max="50"
              value={session.rider}
              onChange={(e) =>
                void act({
                  type: 'rider',
                  value: Number(e.target.value),
                })
              }
            />
          </label>
          <button
            className="text-button"
            onClick={() => void act({ type: 'rider', value: 0 })}
          >
            游码归零
          </button>
        </div>
      </div>
      <div className="record-row">
        <label htmlFor="reading"> 测量读数</label>
        <div className="unit-input">
          <input
            id="reading"
            aria-label=" 测量读数"
            type="number"
            min="0"
            max="1000"
            step="0.1"
            placeholder="0.0"
            value={reading}
            onChange={(e) => setReading(e.target.value)}
          />
          <span>g</span>
        </div>
        <button
          className="primary"
          disabled={reading === '' || !Number.isFinite(Number(reading))}
          onClick={submitReading}
        >
          记录读数
```

```
                    </button>
                    <button
                      className="secondary"
                      onClick={() => void act({ type: 'tidy' })}
                    >
                      整理器材归位
                    </button>
                  </div>
                </fieldset>
              </section>
            </div>
            <aside>
              <section className="assessment">
                <div className="panel-heading">
                  <strong> 实验完成度</strong>
                  <span className="subtle">8 个环节</span>
                </div>
                <div className="score">
                  <strong>{session.score}</strong>
                  <span>/ 100 分</span>
                  <p> 练习完成评分</p>
                </div>
                <div className="progress">
                  <div style={{ width: session.score + '%' }} />
                </div>
                <ol className="step-list">
                  {session.steps.map((s, i) => (
                    <li key={s.name} className={s.passed ? 'passed' : ''}>
                      <span className="step-no">
                        {s.passed ? '✓' : String(i + 1).padStart(2, '0')}
                      </span>
                      <span>{s.name}</span>
                      <b>
                        {s.score}
                        <small>/{s.max}</small>
                      </b>
                    </li>
                  ))}
                </ol>
                <p className="assessment-note">
                  操作失误会记录提示；纠正后仍可完成环节得分。每项只计分一次。
                </p>
                <button
                  className="primary full"
                  disabled={busy}
                  onClick={() =>
                    session.status === 'finished'
                      ? setReport(session)
                      : void finish()
                  }
```

```
                  >
                    {session.status === 'finished'
                      ? '查看实验报告'
                      : '结束实验 · 生成报告'}
                  </button>
                </section>
                <section className="event-panel">
                  <div className="panel-heading">
                    <strong> 最近操作</strong>
                    <span className="subtle"> 自动保存</span>
                  </div>
                  {session.events
                    .slice(-3)
                    .reverse()
                    .map((e, i) => (
                      <div className="event" key={i}>
                        <i className={e.violation ? 'warn' : ''} />
                        <p>
                          {e.message}
                          <time>
                            {new Date(e.at).toLocaleTimeString('zh-CN', {
                              hour12: false,
                            })}
                          </time>
                        </p>
                      </div>
                    ))}
                  {!session.events.length && (
                    <p className="subtle"> 从检查实验器材开始。</p>
                  )}
                </section>
              </aside>
            </div>
          )}
        </>
      )}
      {tab === '实验记录' && (
        <>
          <div className="page-heading">
            <div className="eyebrow"> 实验档案 / 自动保存</div>
            <h1> 每一次练习，都有迹可循。</h1>
            <p> 查看测量结果、操作过程与分项完成情况。</p>
          </div>
          <section className="history-panel">
            <table>
              <thead>
                <tr>
                  <th> 实验名称</th>
                  <th> 创建时间</th>
                  <th> 来源</th>
```

```
                  <th> 状态</th>
                  <th> 完成评分</th>
                  <th> 操作</th>
                </tr>
              </thead>
              <tbody>
                {history.map((s) => (
                  <tr key={s.id}>
                    <td>
                      <strong>{s.label}</strong>
                      <small>{s.id.slice(0, 8)}</small>
                    </td>
                    <td>{date(s.createdAt)}</td>
                    <td>{s.source === 'demo' ? '演示记录' : '自主练习'}</td>
                    <td>
                      <span className="badge">
                        {s.status === 'finished' ? '已归档' : '进行中'}
                      </span>
                    </td>
                    <td>
                      <b>{s.score}</b> / 100
                    </td>
                    <td>
                      <button className="text-button" onClick={() => setReport(s)}>
                        查看报告
                      </button>
                      <button className="text-button" onClick={() => download(s)}>
                        导出 JSON
                      </button>
                    </td>
                  </tr>
                ))}
              </tbody>
            </table>
            {!history.length && (
              <div className="empty"> 还没有实验记录。前往实验台开始练习。</div>
            )}
          </section>
        </>
      )}
      {tab === '实验指南' && (
        <>
          <div className="page-heading">
            <div className="eyebrow"> 操作规范 / 学习指南</div>
            <h1> 先调平，再测量。</h1>
            <p> 这是一套本地交互练习系统，八个环节共 100 分。</p>
          </div>
          <div className="guide-grid">
            {STEP_NAMES.map((name, i) => (
              <section key={name}>
```

```
                <span className="guide-no">{String(i + 1).padStart(2, '0')}</span>
                <h2>
                  {name}
                  <small>{STEP_POINTS[i]} 分</small>
                </h2>
                <p>
                  {
                    [
                      '先检查天平、砝码、镊子及待测物体，确认器材齐全。',
                      '保持托盘空载，将游码移动到标尺的零刻度。',
                      '游码归零后，将平衡螺母调整到 0，使空载天平平衡。',
                      '完成调平后，将待测物体放在左盘，砝码放在右盘。',
                      '用镊子先试较大的砝码，过重则取回，再依次试较小的砝码。',
                      '砝码接近物体质量后，微调游码；指针居中即表示达到平衡。',
                      '物体质量等于右盘砝码质量加上游码示值，按 0.1 g 分度记录。',
                      '记录正确读数后整理器材：物体、砝码归盒，游码与螺母回零。',
                    ][i]
                  }
                </p>
              </section>
            ))}
          </div>
          <p className="guide-disclaimer">
            评分用于练习完成情况反馈，不能替代真实实验考核。错误操作保留在报告中，纠正后可继续完成练习。
          </p>
        </>
      )}
    </main>
    <footer>
      <span>
        {SOFTWARE_NAME} {VERSION}
      </span>
      <span> 本地运行 · 过程留痕 · 自主练习</span>
    </footer>
    {reset && (
      <div className="modal-backdrop">
        <section
          className="new-dialog"
          role="dialog"
          aria-modal="true"
          aria-labelledby="new-title"
        >
          <h2 id="new-title"> 新建一轮实验</h2>
          <p> 当前记录会保留在实验记录中，新实验从初始状态开始。</p>
          <label>
            实验名称
            <input
              value={label}
              onChange={(e) => setLabel(e.target.value)}
              maxLength={80}
```

```
            />
          </label>
          <label>
            记录来源
            <select
              value={source}
              onChange={(e) => setSource(e.target.value as 'practice' | 'demo')}
            >
              <option value="practice"> 自主练习</option>
              <option value="demo"> 演示记录</option>
            </select>
          </label>
          <div className="dialog-actions">
            <button className="secondary" onClick={() => setReset(false)}>
              取消
            </button>
            <button className="primary" disabled={busy} onClick={() => void create()}>
              开始新实验
            </button>
          </div>
        </section>
      </div>
    )}
    {report && (
      <div className="modal-backdrop report-backdrop">
        <section
          className="report"
          role="dialog"
          aria-modal="true"
          aria-labelledby="report-title"
        >
          <div className="report-tools">
            <button className="secondary" onClick={() => download(report)}>
              导出 JSON
            </button>
            <button className="primary" onClick={() => window.print()}>
              打印 / 保存 PDF
            </button>
            <button className="secondary" onClick={() => setReport(null)}>
              关闭报告
            </button>
          </div>
          <div className="eyebrow">
            实验评测报告 · {report.source === 'demo' ? '演示记录' : '自主练习'}
          </div>
          <h1 id="report-title">{report.label}</h1>
          <p>
            {SOFTWARE_NAME} {VERSION}
          </p>
          <div className="report-summary">
```

```
            <div>
              <span> 完成评分</span>
              <strong>{report.score} / 100</strong>
            </div>
            <div>
              <span> 记录读数</span>
              <strong>
                {report.reading === null ? '尚未记录' : grams(report.reading) + ' g'}
              </strong>
            </div>
            <div>
              <span> 状态</span>
              <strong>{report.status === 'finished' ? '已归档' : '进行中'}</strong>
            </div>
          </div>
          <p className="subtle">
            创建时间：{date(report.createdAt)}
            <br />
            结束时间：{report.finishedAt ? date(report.finishedAt) : '尚未结束'}
            <br />
            记录编号：{report.id}
          </p>
          <h2> 测量结果</h2>
          <p>
            {report.measurement
              ? `右盘砝码 ${grams(report.measurement.weightTotal)} g ＋ 游码
              ${grams(report.measurement.rider)} g ＝ 物体质量 ${grams(report.measurement.value)} g`
              : '尚未完成正确读数。'}
          </p>
          <h2> 分项完成情况</h2>
          <table>
            <thead>
              <tr>
                <th> 实验环节</th>
                <th> 状态</th>
                <th> 得分</th>
              </tr>
            </thead>
            <tbody>
              {report.steps.map((s) => (
                <tr key={s.name}>
                  <td>{s.name}</td>
                  <td>{s.passed ? '已完成' : '未完成'}</td>
                  <td>
                    {s.score} / {s.max}
                  </td>
                </tr>
              ))}
            </tbody>
          </table>
          <h2> 完整操作记录</h2>
```

```
            <table className="report-events">
              <thead>
                <tr>
                  <th> 时间</th>
                  <th> 类型</th>
                  <th> 操作反馈</th>
                </tr>
              </thead>
              <tbody>
                {report.events.map((e, i) => (
                  <tr key={i}>
                    <td>{date(e.at)}</td>
                    <td>{e.violation ? '违规提示' : '操作记录'}</td>
                    <td>{e.message}</td>
                  </tr>
                ))}
              </tbody>
            </table>
            <p className="report-note">
              本报告为本地交互练习的完成评分。失误记录保留，纠正后允许获得环节得分；报告不代替真实实验考核。
            </p>
          </section>
        </div>
      )}
    </>
  );
}
createRoot(document.getElementById('root')!).render(
  <React.StrictMode>
    <App />
  </React.StrictMode>,
);
:root {
  font-family: 'Microsoft YaHei', 'PingFang SC', sans-serif;
  color: #243e4e;
  background: #f3f6f8;
  font-synthesis: none;
  font-size: 14px;
  --navy: #163c51;
  --teal: #14857d;
  --line: #e0e8ec;
  --muted: #718592;
}
* {
  box-sizing: border-box;
}
body {
  margin: 0;
}
button,
```

```
input,
select {
  font: inherit;
}
button {
  cursor: pointer;
  transition:
    background 0.15s,
    box-shadow 0.15s;
}
button:disabled {
  cursor: not-allowed;
  opacity: 0.48;
}
button:focus-visible,
input:focus-visible,
select:focus-visible {
  outline: 3px solid #55b5bf;
  outline-offset: 3px;
}
button {
  border: 0;
  border-radius: 5px;
}
input,
select {
  border: 1px solid #d7e1e6;
  background: white;
  color: #284555;
  border-radius: 4px;
  padding: 9px 10px;
  min-width: 0;
}
input[type='range'] {
  width: 100%;
  accent-color: var(--teal);
  padding: 0;
  height: 20px;
  cursor: pointer;
}
input[type='range']:disabled {
  cursor: not-allowed;
}
.topbar {
  height: 81px;
  background: #fff;
  display: flex;
  align-items: center;
  padding: 0 4%;
  gap: 14px;
```

```
  border-bottom: 1px solid var(--line);
}
.brand-icon {
  background: var(--navy);
  color: white;
  width: 43px;
  height: 43px;
  display: grid;
  place-items: center;
  border-radius: 8px;
  font-size: 27px;
}
.brand {
  display: flex;
  flex-direction: column;
  gap: 5px;
}
.brand strong {
  font-size: 17px;
  letter-spacing: 0.5px;
}
.brand span {
  color: var(--muted);
  font-size: 11px;
  letter-spacing: 2px;
}
.version {
  font-family: Consolas, monospace;
  font-size: 11px;
  background: #eef3f6;
  color: #758c99;
  padding: 4px 7px;
  border-radius: 4px;
  align-self: center;
}
.connection {
  margin-left: auto;
  display: flex;
  align-items: center;
  gap: 7px;
  color: #728391;
  font-size: 12px;
}
.connection i {
  width: 6px;
  height: 6px;
  background: #a5b1ba;
  border-radius: 50%;
}
.connection i.on {
```

```
  background: #159583;
  box-shadow: 0 0 0 4px #edf7f4;
}
.navigation {
  height: 55px;
  background: white;
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding: 0 4%;
  border-bottom: 1px solid var(--line);
}
.navigation nav {
  display: flex;
  height: 100%;
  gap: 30px;
}
.navigation button {
  padding: 0 5px;
  background: transparent;
  border-radius: 0;
  color: #748590;
  border-bottom: 3px solid transparent;
  display: flex;
  align-items: center;
  gap: 10px;
}
.navigation button span {
  font-size: 18px;
}
.navigation button.selected {
  color: var(--teal);
  border-bottom-color: var(--teal);
  font-weight: 700;
}
.navigation > span {
  font-size: 11px;
  color: #91a1aa;
}
main {
  max-width: 1510px;
  margin: auto;
  padding: 25px 4% 35px;
  min-height: calc(100vh - 181px);
}
.page-heading {
  position: relative;
  margin: 0 0 23px;
}
.page-heading > button {
```

```
  position: absolute;
  right: 0;
  bottom: 2px;
}
.eyebrow {
  font-size: 11px;
  letter-spacing: 2px;
  color: #648995;
  font-weight: 600;
  margin-bottom: 8px;
}
h1 {
  font-size: 26px;
  letter-spacing: 1px;
  color: #1b3c50;
  font-weight: 600;
  margin: 0 0 8px;
}
p {
  line-height: 1.7;
}
h2 {
  font-size: 18px;
}
h3 {
  font-size: 12px;
  font-weight: 600;
  margin: 0 0 14px;
}
.page-heading p {
  font-size: 12px;
  color: var(--muted);
  margin: 0;
}
.primary {
  padding: 10px 16px;
  background: var(--teal);
  color: white;
  font-weight: 600;
  border: 1px solid var(--teal);
}
.primary:hover:not(:disabled) {
  background: #0c716c;
}
.secondary {
  background: white;
  border: 1px solid #d8e3e8;
  padding: 10px 15px;
  color: #446372;
}
```

```
.secondary:hover:not(:disabled) {
  background: #eef5f7;
}
.text-button {
  background: transparent;
  color: var(--teal);
  padding: 6px;
  font-size: 12px;
}
.workbench {
  display: grid;
  grid-template-columns: minmax(0, 1fr) 310px;
  gap: 22px;
  align-items: start;
}
.left-column {
  min-width: 0;
}
.instrument-panel,
.controls,
.assessment,
.event-panel,
.history-panel,
.guide-grid section {
  background: white;
  border: 1px solid var(--line);
  border-radius: 8px;
  overflow: hidden;
}
.panel-heading {
  padding: 17px 20px;
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 10px;
}
.panel-heading strong {
  font-size: 13px;
  font-weight: 600;
}
.panel-heading > div {
  display: flex;
  align-items: center;
  gap: 9px;
}
.badge {
  font-size: 10px;
  background: #edf5f4;
  color: #467d78;
  padding: 4px 7px;
```

```
  border-radius: 3px;
  white-space: nowrap;
}
.subtle {
  color: var(--muted);
  font-size: 11px;
}
.instrument-panel > .panel-heading {
  background: #faffff;
  border-bottom: 1px solid var(--line);
}
.balance {
  width: 100%;
  height: auto;
  max-height: 330px;
  display: block;
  background-color: #f0f7fa;
  background-image:
    linear-gradient(#dfebf080 1px, transparent 1px),
    linear-gradient(90deg, #dfebf080 1px, transparent 1px);
  background-size: 24px 24px;
}
.pan-label {
  font-size: 12px;
  fill: #73909e;
  letter-spacing: 2px;
}
.beam {
  transition: transform 0.3s ease-out;
}
.instrument-readings {
  display: flex;
  border-top: 1px solid var(--line);
  padding: 15px 0;
  background: #fff;
}
.instrument-readings > div {
  flex: 1;
  display: flex;
  align-items: center;
  justify-content: center;
  gap: 15px;
  border-right: 1px solid var(--line);
}
.instrument-readings > div:last-child {
  border: 0;
}
.instrument-readings span {
  font-size: 11px;
  color: var(--muted);
```

```
}
.instrument-readings strong {
  font-size: 23px;
  font-family: Consolas, monospace;
  font-weight: 500;
  color: #234e63;
}
.instrument-readings small {
  font-size: 11px;
  margin-left: 5px;
  color: var(--muted);
}
.instrument-readings .status-word {
  font-family: inherit;
  font-size: 13px;
  color: var(--teal);
}
.feedback {
  display: flex;
  gap: 11px;
  background: #e7f3f2;
  border: 1px solid #d2e8e4;
  border-radius: 5px;
  padding: 12px 16px;
  margin: 14px 0;
  color: #40766e;
}
.feedback > span {
  font:
    600 11px Georgia,
    serif;
  width: 18px;
  height: 18px;
  border: 1px solid #75a79f;
  border-radius: 50%;
  text-align: center;
  line-height: 17px;
  flex-shrink: 0;
  margin-top: 1px;
}
.feedback strong {
  font-size: 11px;
}
.feedback p {
  font-size: 12px;
  margin: 3px 0 0;
}
.feedback.warning {
  background: #fff5e8;
  border-color: #efddc3;
```

```
  color: #916634;
}
.controls .panel-heading {
  padding-bottom: 12px;
}
.controls fieldset {
  border: 0;
  margin: 0;
  padding: 0;
  min-width: 0;
}
.control-grid {
  display: grid;
  grid-template-columns: 1fr 1fr 1.15fr;
  padding: 5px 20px 18px;
  gap: 18px;
}
.control-group {
  border-right: 1px solid var(--line);
  padding-right: 18px;
}
.control-group:last-child {
  border: 0;
  padding: 0;
}
.control-group h3 b {
  color: #8babb7;
  font:
    11px Consolas,
    monospace;
  margin-right: 4px;
}
.control-group > button.secondary,
.done-button {
  padding: 8px 10px;
  width: 100%;
  font-size: 11px;
}
.done-button {
  background: #edf6f3;
  color: #367b6e;
  border: 1px solid #d7e9e1;
}
.control-group label {
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 7px;
  font-size: 11px;
  color: #708691;
```

```
  margin: 8px 0;
}
.control-group select {
  font-size: 11px;
  padding: 5px 3px;
  max-width: 64%;
  width: 125px;
}
.slider-label {
  flex-wrap: wrap;
  margin-top: 13px !important;
}
.slider-label output {
  font-family: Consolas, monospace;
  color: #335c6c;
  font-size: 12px;
}
.control-group > small {
  font-size: 10px;
  color: #91a0a8;
  line-height: 1.5;
  display: block;
}
.weight-buttons {
  display: flex;
  gap: 6px;
}
.weight-buttons button {
  flex: 1;
  background: #f5f3eb;
  border: 1px solid #e5dfcb;
  border-radius: 5px;
  padding: 7px 3px;
  color: #7c6b49;
  font:
    14px Consolas,
    monospace;
}
.weight-buttons button span {
  display: block;
  font-size: 18px;
  line-height: 18px;
  color: #a28d5d;
}
.weight-buttons button small {
  font-size: 9px;
  margin-left: 2px;
}
.placed-weights {
  display: flex;
```

```
  gap: 4px;
  flex-wrap: wrap;
  min-height: 23px;
  padding-top: 6px;
}
.placed-weights button {
  font-size: 10px;
  background: #ecf2f5;
  color: #668492;
  padding: 3px 5px;
}
.placed-weights > span {
  font-size: 10px;
  color: #94a2aa;
}
.record-row {
  border-top: 1px solid var(--line);
  padding: 15px 20px;
  display: flex;
  align-items: center;
  gap: 10px;
  background: #fbfcfd;
}
.record-row label {
  font-size: 12px;
  color: #5d7988;
  margin-right: 4px;
  white-space: nowrap;
}
.unit-input {
  display: flex;
  border: 1px solid #d7e1e6;
  background: white;
  border-radius: 4px;
  align-items: center;
  width: 135px;
}
.unit-input input {
  border: 0;
  width: 100%;
  padding: 8px;
  font-family: Consolas, monospace;
}
.unit-input span {
  padding-right: 10px;
  font-size: 11px;
  color: #91a1a9;
}
.record-row button {
  font-size: 11px;
```

```
  padding: 9px 12px;
  white-space: nowrap;
}
.record-row > button:last-child {
  margin-left: auto;
}
.assessment {
  padding-bottom: 18px;
}
.assessment > .panel-heading {
  border-bottom: 1px solid var(--line);
}
.score {
  text-align: center;
  padding: 20px 0 4px;
}
.score strong {
  font:
    48px 'Bahnschrift',
    Consolas,
    monospace;
  color: var(--navy);
}
.score > span {
  font-size: 12px;
  color: #8ca0aa;
  margin-left: 5px;
}
.score p {
  font-size: 11px;
  color: #88a0aa;
  letter-spacing: 2px;
  margin: 3px;
}
.progress {
  height: 4px;
  background: #edf2f4;
  margin: 13px 23px 15px;
  overflow: hidden;
  border-radius: 4px;
}
.progress > div {
  height: 100%;
  background: var(--teal);
  transition: width 0.25s;
}
.step-list {
  list-style: none;
  margin: 0;
  padding: 0 21px;
```

```
}
.step-list li {
  display: flex;
  align-items: center;
  gap: 9px;
  border-bottom: 1px solid #eff3f5;
  padding: 13px 0;
  font-size: 11px;
  color: #637f8c;
}
.step-list li:last-child {
  border-bottom: 0;
}
.step-no {
  border: 1px solid #dce6ea;
  color: #96a9b2;
  font:
    10px Consolas,
    monospace;
  width: 24px;
  height: 24px;
  border-radius: 50%;
  display: grid;
  place-items: center;
  flex-shrink: 0;
}
.step-list li.passed .step-no {
  background: #e3f2ed;
  border-color: #e3f2ed;
  color: #1c8b75;
}
.step-list li.passed > span:nth-child(2) {
  color: #286c61;
}
.step-list li b {
  margin-left: auto;
  color: #4b7082;
  font:
    14px Consolas,
    monospace;
  white-space: nowrap;
}
.step-list li small {
  font-size: 10px;
  color: #9bafb8;
}
.assessment-note {
  padding: 0 21px;
  color: #94a5ad;
  font-size: 10px;
```

```
  margin: 13px 0 17px;
  line-height: 1.8;
}
.full {
  display: block;
  margin: 0 20px;
  width: calc(100% - 40px);
  font-size: 12px;
}
.event-panel {
  margin-top: 15px;
  padding: 0 19px 10px;
}
.event-panel .panel-heading {
  padding: 16px 0 8px;
}
.event {
  display: flex;
  gap: 10px;
  font-size: 10px;
  color: #7b8f99;
}
.event > i {
  width: 5px;
  height: 5px;
  border-radius: 50%;
  background: #88b1aa;
  flex-shrink: 0;
  margin-top: 14px;
}
.event > i.warn {
  background: #d0a26c;
}
.event p {
  margin: 7px 0;
  line-height: 1.6;
}
.event time {
  display: block;
  font-family: Consolas, monospace;
  font-size: 9px;
  color: #a3b0b7;
  margin-top: 3px;
}
footer {
  height: 45px;
  display: flex;
  align-items: center;
  justify-content: space-between;
  padding: 0 4%;
```

```
  font-size: 10px;
  color: #96a7b1;
  border-top: 1px solid #e3eaee;
  background: #f7f9fa;
}
.start-panel {
  background: #fff;
  border: 1px solid var(--line);
  padding: 42px;
  max-width: 590px;
  margin: 35px auto;
  border-radius: 8px;
  display: flex;
  flex-direction: column;
  gap: 13px;
}
.start-panel h2,
.start-panel p {
  margin: 0;
  text-align: center;
}
.start-panel p {
  font-size: 12px;
  color: var(--muted);
  margin-bottom: 13px;
}
.start-mark {
  text-align: center;
  font-size: 55px;
  color: #356979;
}
.start-panel label,
.new-dialog label {
  display: flex;
  flex-direction: column;
  gap: 8px;
  color: #78909d;
  font-size: 12px;
}
.start-panel .primary {
  margin-top: 9px;
}
.history-panel {
  overflow: auto;
}
table {
  width: 100%;
  border-collapse: collapse;
  text-align: left;
  font-size: 12px;
```

```
}
th {
  background: #eef4f6;
  padding: 15px 20px;
  color: #658392;
  font-weight: 500;
  white-space: nowrap;
}
td {
  padding: 18px 20px;
  border-bottom: 1px solid #e8eef1;
}
td small {
  display: block;
  font:
    10px Consolas,
    monospace;
  color: #9aadb7;
  margin-top: 6px;
}
td strong {
  font-weight: 500;
}
.empty {
  text-align: center;
  padding: 75px 20px;
  color: #7a929e;
}
.guide-grid {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  gap: 18px;
}
.guide-grid section {
  padding: 25px;
}
.guide-no {
  font:
    36px 'Bahnschrift',
    Consolas,
    monospace;
  color: #a7c4cf;
}
.guide-grid h2 {
  font-size: 14px;
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-top: 20px;
}
```

```
.guide-grid h2 small {
  font-size: 10px;
  color: #789eaa;
  font-weight: 400;
}
.guide-grid p {
  font-size: 12px;
  color: #78909d;
}
.guide-disclaimer {
  font-size: 12px;
  color: #78909d;
  margin-top: 25px;
}
.error {
  background: #fff0e9;
  color: #9f5a37;
  padding: 12px;
  border: 1px solid #e7ccb7;
  margin-bottom: 20px;
  border-radius: 5px;
  font-size: 12px;
}
.error button {
  margin-left: 15px;
  color: #945632;
  background: white;
  padding: 5px;
}
.modal-backdrop {
  position: fixed;
  inset: 0;
  background: #15344466;
  backdrop-filter: blur(3px);
  display: flex;
  align-items: center;
  justify-content: center;
  z-index: 5;
  padding: 25px;
}
.new-dialog {
  background: white;
  padding: 30px;
  border-radius: 10px;
  width: 450px;
  max-width: 100%;
  box-shadow: 0 20px 80px #122b4133;
}
.new-dialog h2 {
  margin-top: 0;
```

```
}
.new-dialog p {
  font-size: 12px;
  color: #7a929e;
}
.new-dialog label {
  margin-top: 18px;
}
.dialog-actions {
  display: flex;
  justify-content: flex-end;
  gap: 9px;
  margin-top: 25px;
}
.report {
  background: white;
  width: 900px;
  max-height: 90vh;
  overflow: auto;
  padding: 36px;
  border-radius: 9px;
  box-shadow: 0 20px 80px #122b4133;
}
.report-tools {
  display: flex;
  justify-content: flex-end;
  gap: 9px;
  margin-bottom: 30px;
}
.report h1 {
  font-size: 24px;
}
.report > p {
  font-size: 12px;
  color: #8397a2;
}
.report-summary {
  background: #edf5f5;
  display: flex;
  padding: 23px;
  margin: 25px 0;
  gap: 45px;
}
.report-summary div {
  display: flex;
  flex-direction: column;
  gap: 10px;
}
.report-summary span {
  font-size: 11px;
```

```
  color: #78909d;
}
.report-summary strong {
  font:
    22px 'Bahnschrift',
    'Microsoft YaHei',
    sans-serif;
  color: #216563;
}
.report h2 {
  font-size: 15px;
  margin-top: 30px;
}
.report td,
.report th {
  padding: 12px 14px;
  font-size: 11px;
}
.report-events td:first-child {
  white-space: nowrap;
  font-size: 10px;
}
.report-note {
  border-top: 1px solid var(--line);
  margin-top: 25px;
  padding-top: 15px;
}
@media (min-width: 1500px) {
  .balance {
    max-height: 380px;
  }
}
@media (max-width: 1150px) {
  .workbench {
    grid-template-columns: minmax(0, 1fr) 275px;
    gap: 15px;
  }
  .control-grid {
    gap: 10px;
    padding-left: 14px;
    padding-right: 14px;
  }
  .control-group {
    padding-right: 10px;
  }
  .panel-heading {
    padding: 15px;
  }
  .instrument-readings > div {
    gap: 8px;
```

```
  }
  .instrument-readings strong {
    font-size: 20px;
  }
  .record-row {
    padding: 13px;
    gap: 7px;
  }
  .record-row button {
    padding: 9px;
  }
  .unit-input {
    width: 95px;
  }
  .guide-grid {
    grid-template-columns: repeat(2, 1fr);
  }
}
@media (max-width: 850px) {
  .workbench {
    grid-template-columns: 1fr;
  }
  .workbench aside {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 15px;
  }
  .event-panel {
    margin-top: 0;
  }
  .connection {
    font-size: 0;
  }
  .navigation > span {
    display: none;
  }
  .brand strong {
    font-size: 14px;
  }
  .brand span {
    font-size: 9px;
  }
  .topbar {
    gap: 9px;
    padding: 0 20px;
  }
  .page-heading h1 {
    font-size: 23px;
  }
  .page-heading > button {
```

```
    position: static;
    margin-top: 15px;
  }
  main {
    padding: 22px;
  }
  .guide-grid {
    gap: 12px;
  }
}
@media (max-width: 550px) {
  .version {
    display: none;
  }
  .brand-icon {
    width: 34px;
    height: 34px;
    font-size: 22px;
  }
  .brand strong {
    font-size: 12px;
  }
  .brand span {
    letter-spacing: 1px;
  }
  .navigation nav {
    gap: 23px;
  }
  .workbench aside {
    display: block;
  }
  .event-panel {
    margin-top: 15px;
  }
  .control-grid {
    grid-template-columns: 1fr;
  }
  .control-group {
    border-right: 0;
    border-bottom: 1px solid var(--line);
    padding: 10px 0 18px;
  }
  .control-group label {
    font-size: 12px;
  }
  .record-row {
    flex-wrap: wrap;
  }
  .record-row > button:last-child {
    margin-left: 0;
```

```
  }
  .instrument-readings {
    flex-wrap: wrap;
  }
  .instrument-readings > div {
    flex-direction: column;
    gap: 5px;
  }
  .panel-heading > div {
    flex-wrap: wrap;
  }
  .guide-grid {
    grid-template-columns: 1fr;
  }
  .report {
    padding: 20px;
  }
  .report-tools {
    flex-wrap: wrap;
  }
  .report-summary {
    gap: 20px;
    padding: 15px;
  }
  .report-summary strong {
    font-size: 16px;
  }
  footer span:last-child {
    display: none;
  }
  .modal-backdrop {
    padding: 10px;
  }
}
@media (prefers-reduced-motion: reduce) {
  * {
    transition: none !important;
    scroll-behavior: auto !important;
  }
}
@media print {
  @page {
    size: A4;
    margin: 18mm;
  }
  :root,
  html,
  body,
  #root {
    background: #fff;
```

```
}
body:has(.report-backdrop) .topbar,
body:has(.report-backdrop) .navigation,
body:has(.report-backdrop) main,
body:has(.report-backdrop) footer {
  display: none;
}
.report-backdrop {
  position: static;
  display: block;
  background: white;
  padding: 0;
  backdrop-filter: none;
}
.report {
  width: 100%;
  max-height: none;
  overflow: visible;
  box-shadow: none;
  border-radius: 0;
  padding: 0;
}
.report-tools {
  display: none;
}
.report-summary {
  print-color-adjust: exact;
}
.report h2 {
  break-after: avoid;
}
tr {
  break-inside: avoid;
}
.report h1 {
  font-size: 22px;
}
.report td,
.report th {
  padding: 5px 8px;
  font-size: 10px;
}
.report p {
  margin-top: 8px;
  margin-bottom: 8px;
}
.report-events td:first-child {
  font-size: 9px;
}
thead {
```

```
    display: table-header-group;
  }
  .report-note {
    font-size: 10px;
  }
}
<!doctype html><html lang="zh-CN"><head><meta charset="UTF-8"/><meta name="viewport" content="width=device-width,initial-scale=1.0"/><meta name="theme-color" content="#16374b"/><link rel="icon" href="data:image/svg+xml,%3Csvg xmlns='http://www.w3.org/2000/svg' viewBox='0 0 32 32'%3E%3Crect width='32' height='32' rx='6' fill='%2316374b'/%3E%3Ctext x='16' y='24' text-anchor='middle' fill='white' font-size='25'%3E⚖%3C/text%3E%3C/svg%3E"/><title>天平实验训练示例 V1.0</title></head><body><div id="root"></div><script type="module" src="/src/main.tsx"></script></body></html>
import { defineConfig } from 'vite';
import react from '@vitejs/plugin-react';
export default defineConfig({
  plugins: [react()],
  server: { proxy: { '/api': 'http://127.0.0.1:3187' } },
});
$ErrorActionPreference = 'Stop'
$projectRoot = Split-Path -Parent $PSScriptRoot
Set-Location -LiteralPath $projectRoot
$nodeCommand = Get-Command node -ErrorAction SilentlyContinue
if (-not $nodeCommand) { throw 'Node.js is missing. Install Node.js 22.18+ or 24 LTS, then retry.' }
$nodeVersion = [version]((& $nodeCommand.Source --version).TrimStart('v'))
if ($nodeVersion -lt [version]'22.18.0') { throw 'Node.js 22.18 or newer is required.' }
if (-not (Test-Path -LiteralPath (Join-Path $projectRoot 'dist/index.html'))) { throw 'Missing dist/index.html. Run npm ci and npm run build first.' }
$port = 3187
if ($env:PORT) { $port = [int]$env:PORT }
if ($port -lt 1024 -or $port -gt 65535) { throw 'PORT must be 1024..65535.' }
$url = "http://127.0.0.1:$port"
$runtimeDir = Join-Path $projectRoot '.runtime'
New-Item -ItemType Directory -Force -Path $runtimeDir | Out-Null
$probe = New-Object System.Net.Sockets.TcpClient
try { $probe.Connect('127.0.0.1', $port); $occupied = $true } catch { $occupied = $false } finally { $probe.Dispose() }
if ($occupied) { throw "Port $port is already in use. Open $url if this system is running, or stop the occupying process. No process was stopped." }
$serverPath = Join-Path $projectRoot 'server/index.ts'
$process = Start-Process -FilePath $nodeCommand.Source -ArgumentList @('"' + $serverPath + '"') -WorkingDirectory $projectRoot -WindowStyle Hidden -RedirectStandardOutput (Join-Path $runtimeDir 'server.log') -RedirectStandardError (Join-Path $runtimeDir 'server-error.log') -PassThru
@{ pid = $process.Id; serverPath = $serverPath; startedAt = $process.StartTime.ToUniversalTime().ToString('o'); port = $port } | ConvertTo-Json | Set-Content -LiteralPath (Join-Path $runtimeDir 'server.json') -Encoding UTF8
$ready = $false
for ($attempt=0; $attempt -lt 40; $attempt++) {
  Start-Sleep -Milliseconds 250
  try { $health = Invoke-RestMethod -Uri "$url/api/health" -TimeoutSec 1; if ($health.status -eq 'ok') { $ready = $true; break } } catch {}
  if ($process.HasExited) { break }
}
if (-not $ready) { throw "Startup failed. Read .runtime/server-error.log. PID: $($process.Id)" }
Write-Host "Running: $url"
if ($env:BALANCE_NO_BROWSER -ne '1') { Start-Process $url }
$ErrorActionPreference = 'Stop'
$projectRoot = Split-Path -Parent $PSScriptRoot
$pidFile = Join-Path $projectRoot '.runtime/server.json'
if (-not (Test-Path -LiteralPath $pidFile)) { Write-Host 'No recorded server process.'; exit 0 }
$record = Get-Content -LiteralPath $pidFile -Raw -Encoding UTF8 | ConvertFrom-Json
$process = Get-Process -Id $record.pid -ErrorAction SilentlyContinue
if (-not $process) { Write-Host 'Server is already stopped.'; exit 0 }
$command = Get-CimInstance Win32_Process -Filter "ProcessId = $($record.pid)"
$recordStart = [datetime]$record.startedAt
if ($process.ProcessName -ne 'node' -or -not $command.CommandLine.Contains($record.serverPath) -or $process.StartTime.ToUniversalTime().Ticks -ne $recordStart.ToUniversalTime().Ticks) {
  throw 'Process identity changed. Refusing to stop a different process.'
}
Stop-Process -Id $record.pid
Write-Host 'Server stopped. Experiment records are preserved.'
```